

Vincent S. Pronel

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EDUCATION

Bachelor of Science in Computer Science

Kean University

Expected 2027

Union, NJ

- **Relevant Coursework:** Software Engineering*, Analysis of Algorithms*, Computer Architecture*, Object Oriented Analysis & Design, Database Management Systems, Applied Statistics, Matrix & Linear Algebra **in progress, Fall 2026*
- **Leadership & Honors:** Web Developer, WiCyS (Women in Cyber Security) Kean Chapter E-Board | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | GMIS Research Posters Competition | Dean's List, Spring 2024 and Fall 2025

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Java, Python, SQL, PHP, HTML/CSS
- **Frameworks & Libraries:** React, Next.js, Node.js, REST APIs, Tailwind, Framer Motion, GSAP
- **Infrastructure & Databases:** Cloudflare (Workers, D1, R2, Pages), Vercel (CI/CD), Supabase, PostgreSQL, MySQL, Railway
- **Tools & Workflow:** Git/GitHub, Linux (Ubuntu), Claude Code, Codex, MySQL Workbench, Figma, LLM API integration

PROFESSIONAL EXPERIENCE

Independent Web Developer

Jan. 2026 – Present

Freelance Full-Stack Web Development

Sayreville, NJ

- Founded and run a freelance practice shipping production applications for 4 business clients across 4 live sites; own the full lifecycle from requirements gathering with non-technical stakeholders through architecture, implementation, deployment, and ongoing maintenance.
- Engineer each build on **Next.js**, **TypeScript**, and **React** with **PostgreSQL** and Supabase behind API routes; ship serverless backends on **Cloudflare Workers** with D1 and R2, integrate Stripe for payments, and deploy through Vercel CI/CD.
- Live sites: deafsharkcoffee.com (interactive menu, online ordering, staff CMS) | daysicecream.com (4,000+ monthly visitors) | trainingaio.com (booking platform, delivered in 2 weeks) | adzio.io (scroll animations and page transitions with Framer Motion and GSAP).

Computer Science Tutor (Code Samurai)

Jan. 2026 – Present

Kean University

Union, NJ

- Selected by CS faculty for twice-weekly one-on-one sessions with 50+ undergraduates across data structures and algorithms; primarily tutor Database Management Systems, coaching students through relational design and writing efficient **SQL** queries.
- Debug students' own code and queries with them rather than supplying answers, tracing logic step by step so they can reproduce the reasoning independently on later assignments.

Data Visualization & Analytics Intern

July 2026

Group Summer Scholars Research Program, Kean University

Union, NJ

- Lectured daily and mentored 5 high school students under *Dr. Ching-yu Huang*, teaching relational querying and interactive charting with SQL, PHP, MySQL, JavaScript, and Google Charts.
- Authored the curriculum, lesson slides, and troubleshooting guides that took every student from zero experience to a published database-driven website and a research poster presented under my guidance.

RESEARCH PROJECTS

Java Execution Visualizer | *CRA UR2PhD-REU Fellowship*

June 2026 – Present

Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, \$8,000 funded, 400 hours

Kean University

- Developed a full-stack Next.js application that visualizes **Java** program execution, rendering variables, objects, the call stack, and the heap at every executed line; integrated a JDI-based tracer for JDK 21 and built the entire application layer on top of it.
- Designed a JavaParser AST layer that collapses loops and method calls into single playback steps, integrated the Google Gemini API for step explanations, and built the participant harness now running an IRB-approved pilot study.

Jobs vs. Wages | *NSF IUSE Scholars Academy*

May 2026 – Present

NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, \$1,400 funded, 10 weeks

Kean University

- Developed an interactive analytics dashboard in PHP, JavaScript, and Google Charts over 33,000 ZIP-code records aggregated in MySQL into 52 U.S. states and territories; found unemployment explains only 14% of variation in household income while college attainment ($r = 0.79$) is far more predictive.

AI Debate Coach Evaluation | *CRA UR2PhD Research Program*

Feb. 2026 – Apr. 2026

Undergraduate Researcher, Advisor: Dr. Yan Ma

Kean University

- Built the debate application and the full study pipeline and fine-tuned the model configurations under test for a 10-week framework on LLM sycophancy, evaluated by GPT-4-as-a-judge alongside a 12-participant human study.
- Identified that the fine-tuned model fabricated 100% of its cited sources and that only 1 of 12 participants challenged its credibility; humans preferred the prompted model 3 to 1 while the automated judge rated the two comparably.